Experiment 18: Docker Image Push to Docker Hub

1. Lab Objective

Learn how to **publish and distribute container images** using **Docker** by pushing them to **Docker Hub**.

By the end of this lab, participants will:

- Understand the concept of **container registries**
- Tag Docker images correctly
- Push images to **Docker Hub**
- Pull images from the registry

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2. Prerequisites

Before starting the lab ensure:

- **Docker** installed and running
- Docker account created on **Docker Hub**
- Basic Docker knowledge (build, run, images)

Verify Docker installation:

docker --version

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3. Key Concepts

Concept	Description
Docker Registry	Storage location for container images
Docker Hub	Public registry for Docker images
Image Tag	Version identifier for an image
Push	Upload image to registry
Pull	Download image from registry

Common registries used in DevOps:

- **Docker Hub**

- **Azure Container Registry**
- **Amazon Elastic Container Registry**
- **Google Container Registry**

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4. Lab Architecture

Developer Machine

```
    |
  Docker Image
    |
docker push
    ▼
Docker Hub Registry
    |
docker pull
    ▼
Other Systems / Servers
```

This enables **image sharing across teams and environments**.

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5. Step 1 – Create a Docker Hub Account

1. Go to https://hub.docker.com
2. Create an account
3. Note your **Docker Hub username**

Example:

username: devopsdemo

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6. Step 2 – Login to Docker Hub

Authenticate using Docker CLI.

docker login

Enter:

Username

Password

Successful output:

Login Succeeded

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7. Step 3 – Verify Local Docker Images

List existing images.

docker images

Example output:

REPOSITORY TAG IMAGE ID

my-docker-app latest abcd1234

This image was created in the previous experiment.

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8. Step 4 – Tag the Docker Image

Before pushing, the image must include the **Docker Hub username**.

Command:

docker tag my-docker-app username/my-docker-app:1.0

Example:

docker tag my-docker-app devopsdemo/my-docker-app:1.0

Verify the tag:

docker images

Output:

REPOSITORY TAG

devopsdemo/my-docker-app 1.0

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9. Step 5 – Push Image to Docker Hub

Upload the image.

docker push username/my-docker-app:1.0

Example:

docker push devopsdemo/my-docker-app:1.0

Docker will:

1. Compress image layers
2. Upload layers to registry
3. Register image version

Example output:

Pushed

latest: digest: sha256:xxxx

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10. Step 6 – Verify Image in Docker Hub

Open **Docker Hub** dashboard.

Navigate to:

Repositories

You will see:

devopsdemo/my-docker-app

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11. Step 7 – Pull Image from Docker Hub

Test the image distribution.

First remove local image:

docker rmi devopsdemo/my-docker-app:1.0

Now pull the image again.

docker pull devopsdemo/my-docker-app:1.0

Verify:

docker images

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12. Step 8 – Run Container from Docker Hub Image

Run a container from the downloaded image.

```
docker run -d -p 8080:80 devopsdemo/my-docker-app:1.0
```

Open browser:

http://localhost:8080

Your containerized application will run successfully.

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13. Image Tagging Best Practices

DevOps teams use version tags such as:

Tag	Meaning
latest	Most recent build
1.0	Major version
1.1	Minor release
dev	Development build
prod	Production-ready

Example:

myapp:1.0

myapp:1.1

myapp:prod

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14. Demonstration Flow

1️⃣ Build Docker image
2️⃣ Show docker images
3️⃣ Login to Docker Hub
4️⃣ Tag image with username
5️⃣ Push image
6️⃣ Show repository in Docker Hub
7️⃣ Delete local image
8️⃣ Pull image again

This demonstrates **full container image lifecycle**.

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15. DevOps Industry Usage

In real DevOps pipelines:

1. Developer commits code to Git
2. CI pipeline builds Docker image
3. Image pushed to registry
4. Deployment pipeline pulls image

Common tools used:

- **GitHub Actions**
- **Jenkins**
- **Azure DevOps**
- **Kubernetes**

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16. Expected Lab Outcome

After completing this experiment, participants will:

- Understand **container registries**
- Push Docker images to **Docker Hub**
- Pull and run images from a registry
- Enable **image distribution across environments**